

VIBE CODING LAB — SELECTED CHAT HISTORY

Project: Pygame Snake | Repository: https://github.com/psj032005-algo/45_snake

Scope: Important prompts, AI guidance, fixes and test outcomes from Tasks 1-4. This is a curated summary of the conversation, not a verbatim full transcript.

Initial setup and project review

Request (summarized): Review the Vibe Coding lab requirements and the Snake repository, run the original game, then complete the README tasks with separate commits.

AI guidance and result: Set up a Python 3.11 virtual environment for the existing Pygame dependency, ran the game, and recorded a before-change video. The four tasks were handled in sequence.

Task 1 — Collision detection

Prompt (summarized): Analyze the current Snake collision logic and fix unfair collisions, particularly immediate reversal into the snake's body, while keeping normal movement intact.

AI guidance: Update `Snake.set_direction()` to ignore a direction that is the exact opposite of the current direction. Keep existing wall and self-collision checks.

Result: Reversal protection was implemented and tested successfully. Commit: 3d45ffa — Fix snake collision detection.

Task 2 — Game Over screen

Prompt (summarized): Replace the terminal-only Game Over message with a visible in-game Game Over screen and final score.

AI guidance: Update `GameEngine.render()` to draw centered "GAME OVER" and "Final Score" text whenever `game_over` is true.

Result: User confirmed the screen appeared. Commit message: Add game over screen.

Task 3 — Replay and difficulty levels

Key user prompt (verbatim):

"I have completed and tested Task 2 and committed it separately. Now I am working on Task 3 of the Snake game assignment. The task requires adding two features: (1) Replay after Game Over, so the player can restart the game without closing the application, and (2) three difficulty levels — Easy, Medium, and Hard — that change the snake's movement speed. Please analyze the current `main.py`, `game/game_engine.py`, `game/snake.py`, and any relevant existing files. Propose a minimal implementation that fits the current architecture. Keep the existing collision detection and Game Over screen working. Do not add sound effects yet because that belongs to Task 4. Explain which files/methods need to change and how the replay and difficulty selection should work before providing code."

AI guidance: Add `reset()` to restart score, snake, food, and state; use R to replay; provide an initial 1/2/3 difficulty menu with Easy (5), Medium (8), and Hard (12) moves per second.

Important debugging iteration: The initial menu froze because a `continue` skipped `pygame.event.get()`. The event loop was moved before the difficulty-menu rendering, allowing keys 1/2/3 to register.

Result: User confirmed the game starts and works. Commit message: Add replay and difficulty levels.

Task 4 — Sound effects

Prompt (summarized): Add sound when food is eaten and when Game Over occurs, using the available MP3 files, without breaking the previous features.

AI guidance: Initialize pygame.mixer, load sounds/Food.mp3 and sounds/Gameover.mp3 in GameEngine, and call .play() on food collection and on wall/self collisions.

Result: User reported Task 4 done and proceeded to final submission. Commit message: Add sound effects.

Final deliverables and verification

The user reported recording both before-change and after-change gameplay videos, completing the four tasks, and pushing the code. Repository: https://github.com/psj032005-algo/45_snake.

Note: Commit messages beyond Task 1 and the final Git log are described from the conversation; commit hashes and remote status were not independently verified in this document.